

# Q-MAML for HEP

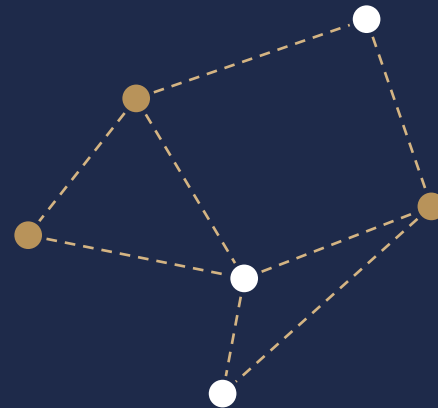
Audit, Replication, and Genuine Hamiltonian Benchmarks

*Ten lattice/field-theory Hamiltonians, eighteen notebooks, one meta-learned VQE initializer*

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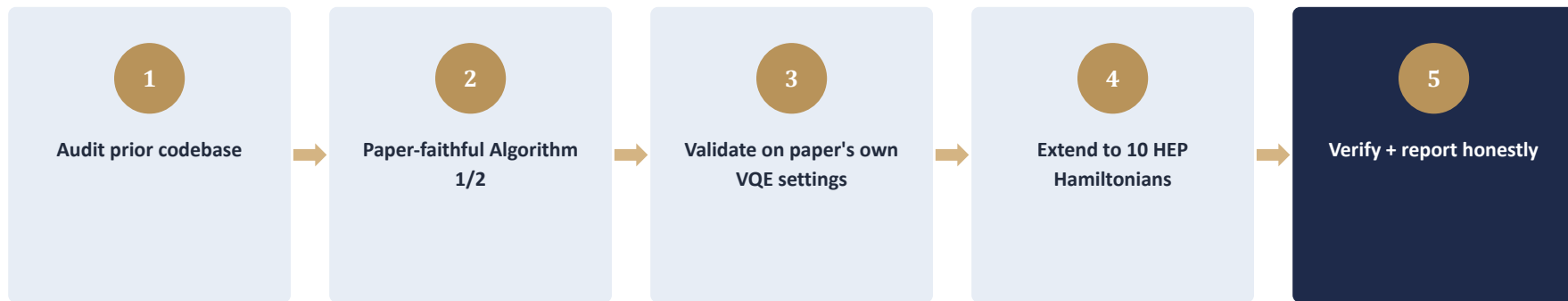
Indian Institute of Science Education and Research Bhopal

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# From Audit to Ten Verified Benchmarks

*Every track follows the same pipeline: build, verify, pretrain, adapt, compare.*



Every track's notebook reports: a verified Hamiltonian, an adaptation-convergence plot, and honest wins, ties, nulls, and losses alike.

# Auditing the Prior Codebase

*Two prior GSoC 2025 attempts to run Q-MAML on HEP classification -- both broken, differently.*



## Quark-gluon pipeline

Gradient detached before backpropagation. The Learner network receives no real training signal. It never actually updates.



## HIGGS pipeline

Runs a first-order-MAML-style inner loop with stop-gradient, not the paper's published, inner-loop-free Algorithm 1.

Fix: a paper-faithful reimplementation, validated against the original paper's own VQE settings (Heisenberg XYZ, H2 molecule) before trusting it on anything new.

# Eighteen Notebooks, Two Parts

*Every Hamiltonian, plot, and real numbers.*

## Part 1 - Foundation (8 notebooks)

- 01 / 01b - HIGGS classification (quick pass + headline)
- 02 / 05 - VQE calibration (Heisenberg XYZ, H2 molecule)
- 03 / 04 - Barren-plateau scans (VQE vs. classification)
- 06 - Inner-step ablation
- 10 - Quark-gluon (2nd classification dataset)

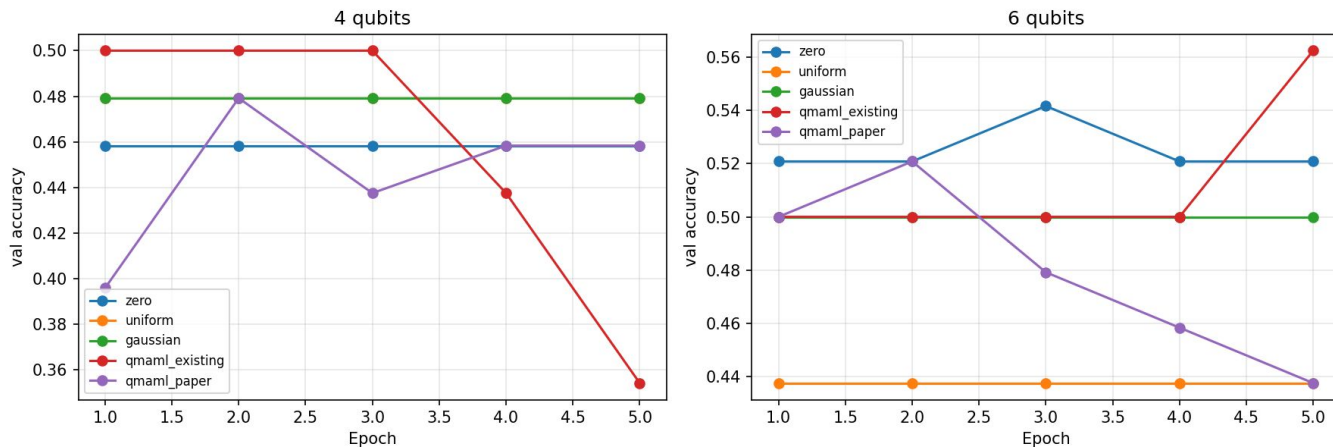
## Part 2 - Genuine HEP Hamiltonians (10 notebooks)

- 07 (A) Schwinger - 08 (B) Z2 LGT
- 09 (C) Scalar field - 12 (D) SU(2) LGT
- 13 (E) Neutrino - 11 (F) SUSY QM
- 14 (G) D8 LGT - 15 (H) Nuclear EFT
- 16 (I) Yukawa - 17 (J) LMG

# HIGGS: Quick Validation Pass

5 epochs, 1 seed, small task pool confirms the harness works end-to-end

Validation accuracy



## 5 epochs

Quick end-to-end pass, not the citable result

## 1 seed

Numbers here are noisy by design

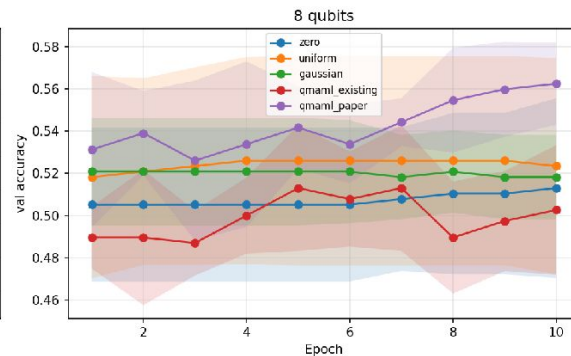
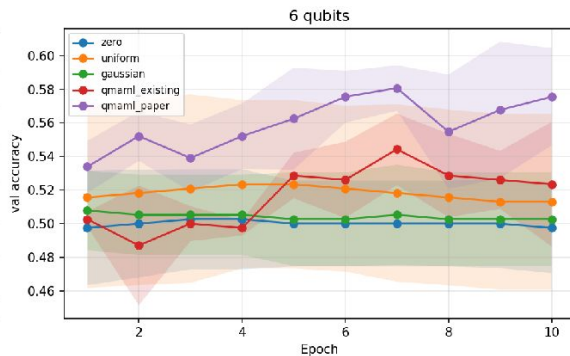
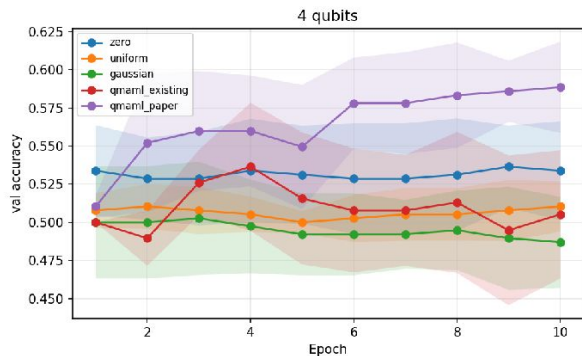
## -> Nb. 01b

Paper-scale, citable version follows next  
this pass just proves the harness works

# HIGGS Classification: The Headline Result

Paper-scale: 10 epochs, 4 seeds, 4 / 6 / 8 qubits

Validation accuracy (mean +/- std over seeds)



**0.589 / 0.576 / 0.563**

Q-MAML accuracy at 4/6/8 qubits - best at every count

**$p < 0.05$**

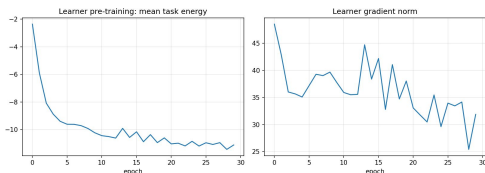
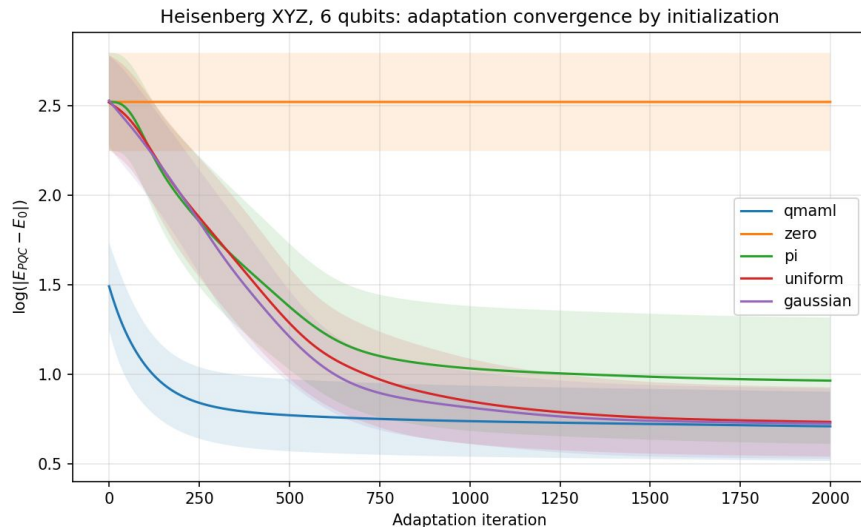
vs. Gaussian init and the GSOC baseline, all 3 qubit counts

**$d = 1.9-3.0$**

Cohen's d vs. Gaussian - a large effect size

# VQE Calibration: Heisenberg XYZ

6 qubits, reconstructed ansatz, 2000 iterations



## Key numbers

**Q-MAML: flat by iter. ~250-500**

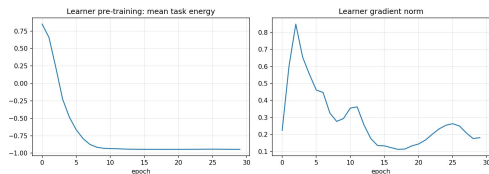
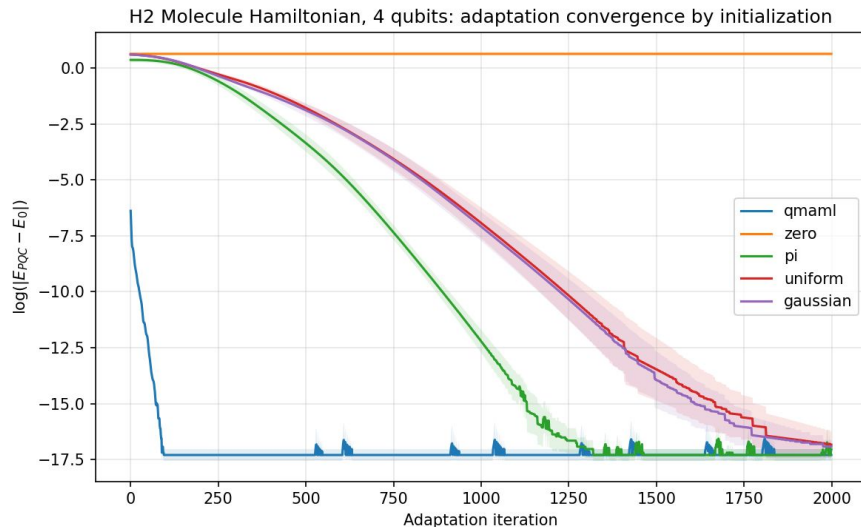
Uniform / Gaussian: need ~1500-2000

Speedup: 4-6x fewer circuit evals

Final log-gaps converge close together across schemes  
the advantage here is convergence speed, not a different final optimum. First calibration check that the reimplemention matches the paper.

# VQE Calibration: H2 Molecule

4 qubits, depth 7, paper's exact ansatz



## Key numbers

**Q-MAML: machine precision by iter. ~100**

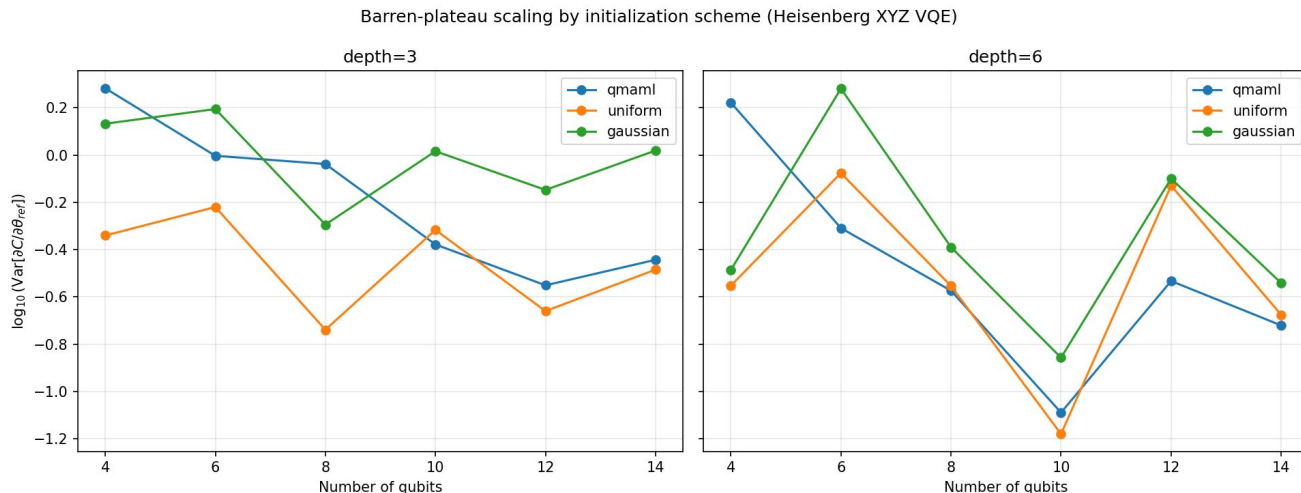
Classical schemes: need ~1500-2000

Speedup: ~15-20x fewer circuit evals

Uses the paper's exact circuit (no reconstruction needed)  
the single strongest piece of evidence that this implementation is faithful to the published algorithm.

# Barren-Plateau Scan: VQE Domain

4-14 qubits, depths 3 & 6, McClean et al. protocol



## No decay

No exponential decay over 4-10 qubits at depth 3

## Comparable

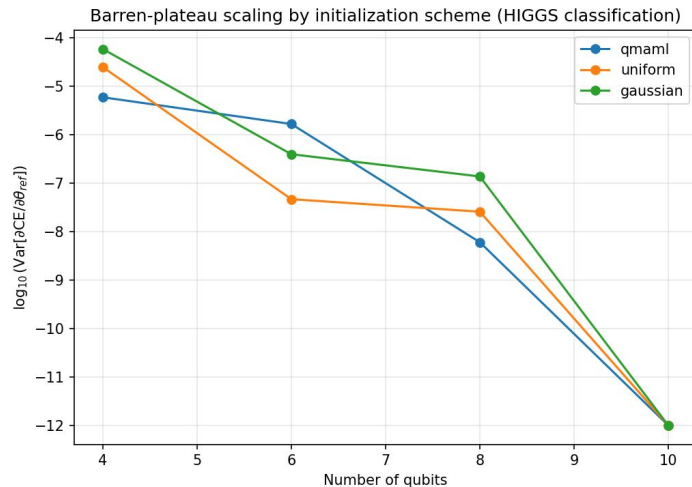
Q-MAML's variance is comparable to or larger than classical

## Placement, not resistance

The VQE advantage is landscape placement

# Barren-Plateau Scan: Classification Domain

HIGGS, the same McClean et al. protocol



**$\sim 10^8 \times$**

Exponential variance decay from 4 to 10 qubits  
a textbook plateau

**Same floor**

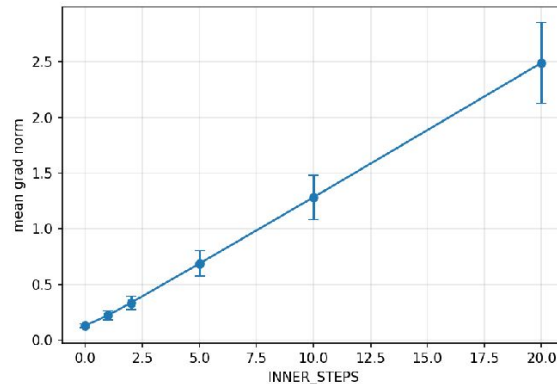
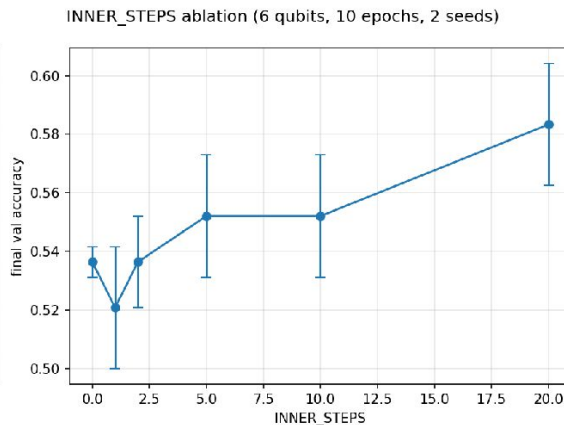
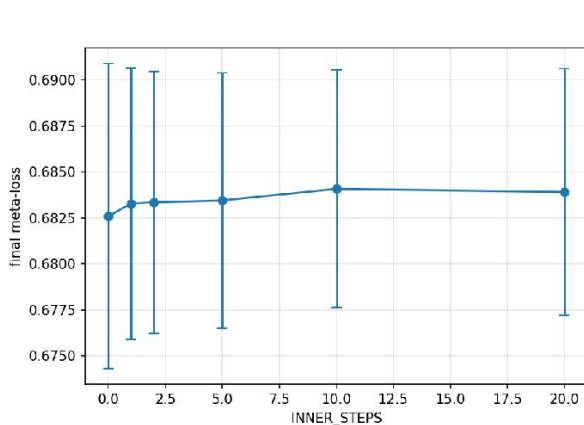
Every scheme collapses to  $\sim 1e-35$  by 10 qubits

**Faster than VQE**

Classification plateaus far faster than the VQE domain

# Inner-Step Ablation

Sweeping INNER\_STEPS 0 -> 20 in the GSoC baseline



**0.536 -> 0.583**

Validation accuracy increases with more inner steps

**0.13 -> 2.49**

Gradient norm scales up, tracking better accuracy

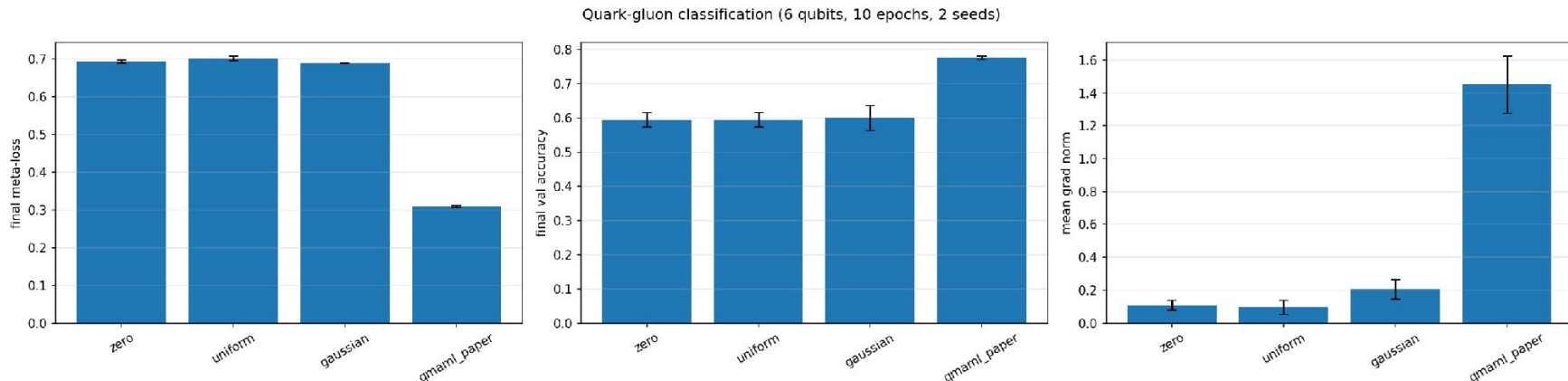
**Retracted**

"More steps = weaker signal" hypothesis does not hold

The original speculative explanation for why Algorithm 1 beats the GSoC baseline is retracted based on this direct test.

## Second Dataset: Quark-Gluon Tagging

6 qubits, 10 epochs, 2 seeds - generalization check



**0.776 vs. ~0.60**

Q-MAML accuracy vs. classical schemes (tightly clustered)

**0.308 vs. ~0.69**

Meta-loss: Q-MAML drops; classical schemes stay at chance level

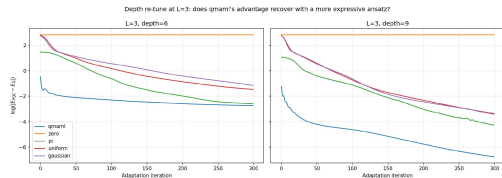
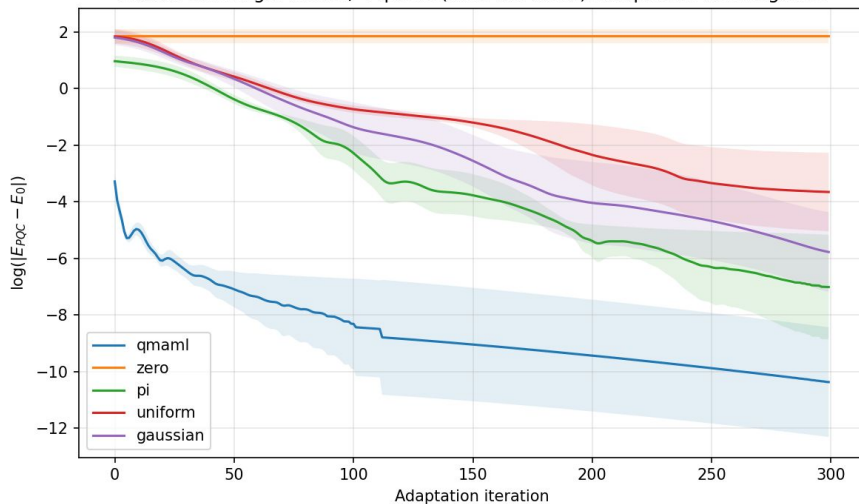
**2 seeds**

Directional replication of the HIGGS win, not yet fully powered

# Lattice Schwinger Model

1+1D QED, gauge field eliminated analytically, 4-8 qubits

Lattice Schwinger model, 4 qubits (held-out tasks): adaptation convergence



## Key numbers

**L=2, Q-MAML: ~0% error, <1% by iter. 4**

L=2, pi / zero: 0.06% / 396%

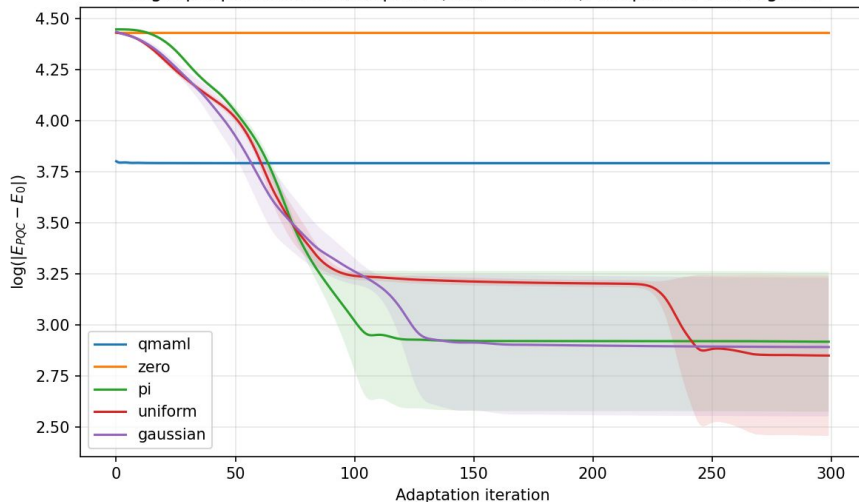
L=3, depth 9: 0.05% (Q-MAML) vs 0.57% (pi)

L=3 at the original depth dips (pi overtakes)  
an ansatz-expressivity artifact, not a breakdown. Re-tuning depth alone recovers and then extends the Q-MAML lead, ~10x better than pi at depth 9.

# Z2 Lattice Gauge Theory

8 qubits, Gauss's law enforced by an energy penalty

Single-plaquette Z2 LGT, 8 qubits (held-out tasks): adaptation convergence



## Key numbers

Uniform (best): 23.07% error

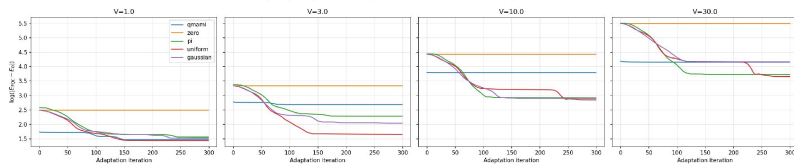
Q-MAML: 59.16% error

3-seed replication: 49.38% +/- 13.70%

The project's one clear loss

the opposite of Track A. A penalty-strength sweep (right) refutes the simple "penalty dominates" explanation without confirming a replacement; the loss itself replicates robustly across seeds.

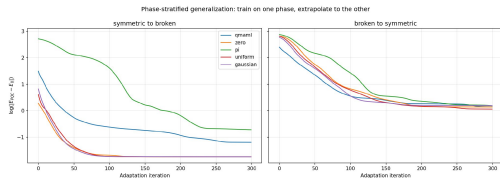
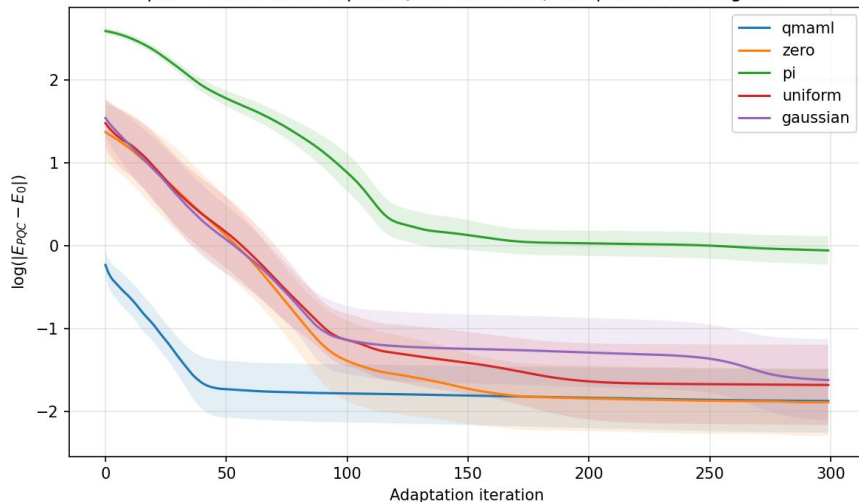
Gauss-law penalty strength sweep: does qmaml recover as V shrinks toward the physical terms' scale?



# $\phi^4$ Scalar Field Theory

6 qubits, bosonic, no gauge structure

$\phi^4$  scalar field, 6 qubits (held-out tasks): adaptation convergence



## Key numbers

**Q-MAML: 5.86% error**

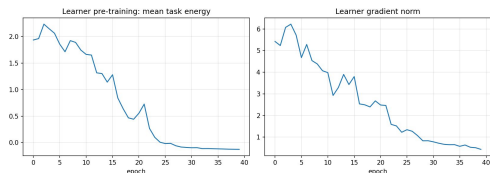
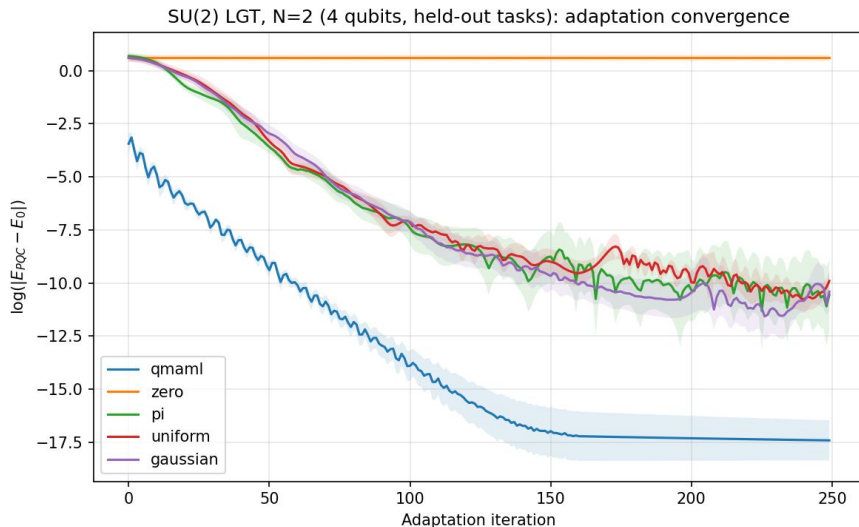
Zero (ties): 5.77% error

pi (worst): 36.06% error

Wins cleanly under interpolation. But the phase-stratified test (right) shows this does not survive extrapolation across the phase transition classical schemes tie or beat Q-MAML out of distribution.

# SU(2) Lattice Gauge Theory

4 qubits,  $N=2$ , non-Abelian gauge eliminated analytically



## Key numbers

**Q-MAML: -17.41 (machine precision)**

Classical schemes: -10 to -11 (plateaued)

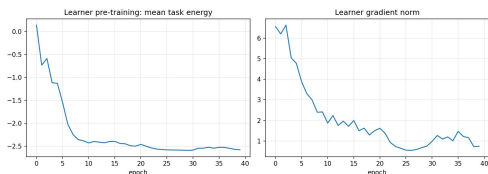
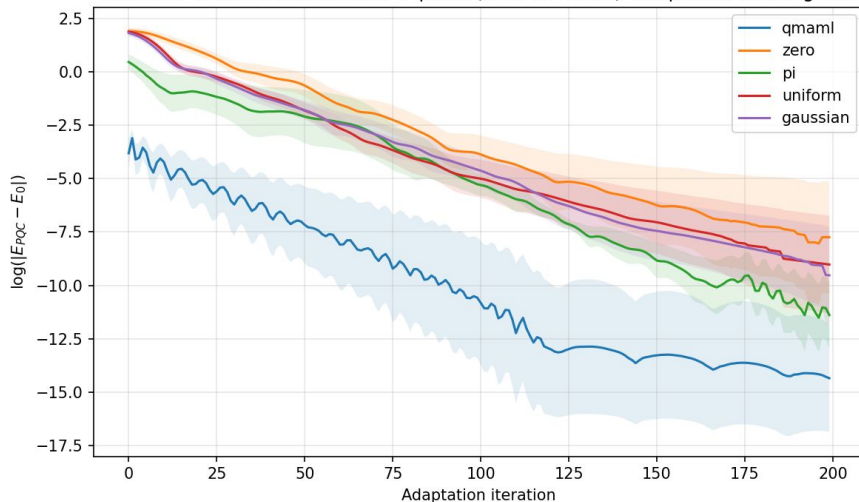
Zero: exact fixed point

The cleanest win in the entire project. Confirms the mechanistic pattern from Track A: gauge structure eliminated analytically (not penalty-enforced) consistently favors Q-MAML.

# Collective Neutrino Oscillations

4 qubits, all-to-all coupling, reformulated as a ground-state problem

Collective neutrino oscillations, 4 qubits (held-out tasks): adaptation convergence



## Key numbers

**Q-MAML: -14.35 (log-gap)**

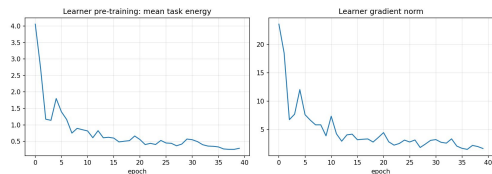
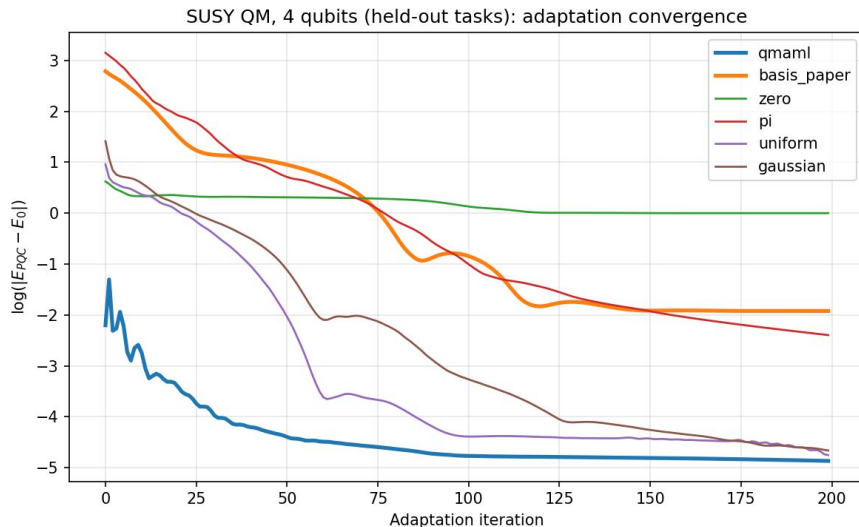
pi (best classical): -11.39

Zero: NOT an exact fixed point

Wins, though not yet converged by iteration 200  
a still-open speed gap. The only Pauli-sum Hamiltonian in the project  
where Zero-init is not an exact fixed point.

# Supersymmetric Quantum Mechanics

4 qubits, discrete superpotential choice + continuous coefficients



## Key numbers

**Q-MAML: -4.87 (log-gap)**

Uniform: -4.75

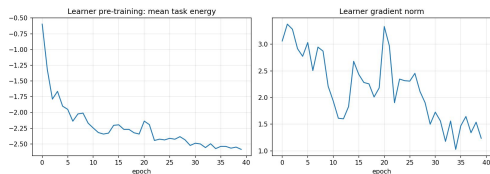
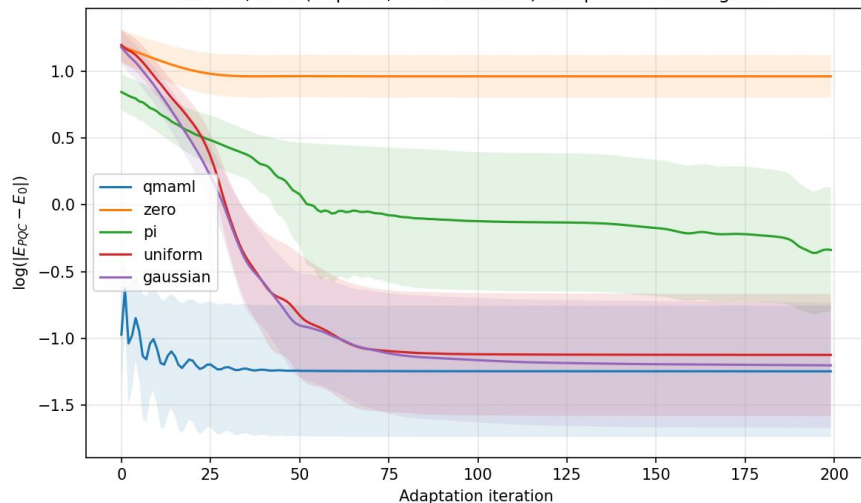
basis\_paper (hand-picked): -1.92 (worst start)

Q-MAML wins. The source paper's own hand-picked, physics-derived initial state is the worst-starting scheme of all six tested fidelity at  $t=0$  is not the same as a good landscape position.

# Non-Abelian D8 Gauge Theory

7 qubits, explicit Jordan-Wigner fermionic matter

D8 LGT, N=2 (7 qubits, held-out tasks): adaptation convergence



## Key numbers

**Q-MAML: 10.49% error**

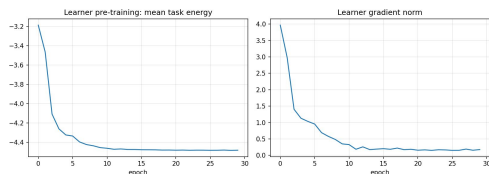
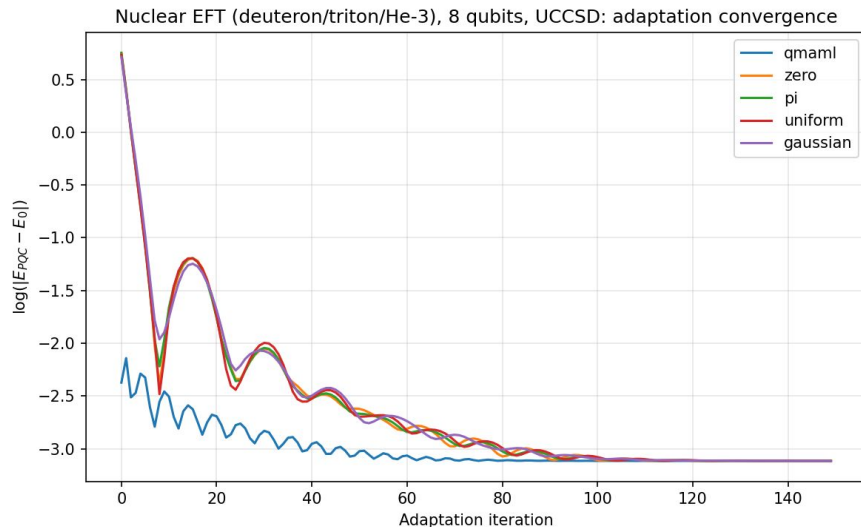
Gaussian: 10.97%

Zero: 95.27% (exact fixed point)

A verification check (64-pair representation-theory test) caught and fixed a real bug a reversed group action before the notebook ever produced a result. Final win is modest but real, clearest in convergence speed.

# Light Nuclei: Deuteron, Triton, He-3

8 qubits, UCCSD ansatz -- the project's only new circuit family



## Key numbers

**Every scheme: 0.89% error (identical)**

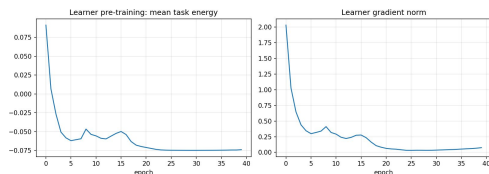
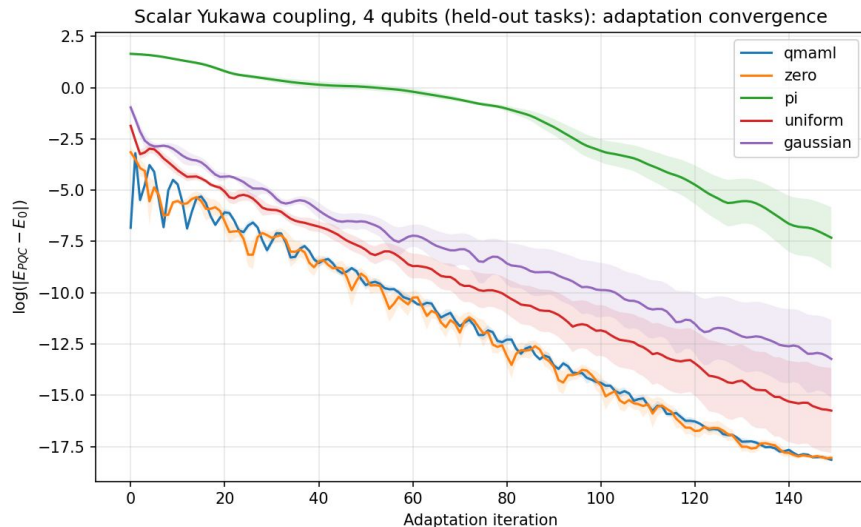
Q-MAML: smooth descent from iter. 0

Classical schemes: shared oscillatory transient

Every scheme reaches the identical final energy  
a scoreboard alone would call this a tie. But Q-MAML descends smoothly while every classical scheme rings through a shared, noisy transient first. A real, explained speed win.

# Scalar Yukawa Coupling

4 qubits, reformulated from real-time dynamics into a ground-state target



## Key numbers

**Q-MAML: -18.15 (precision floor)**

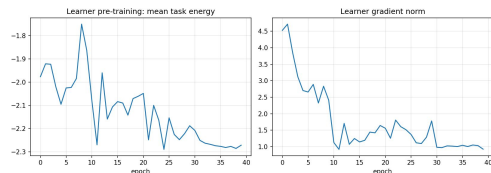
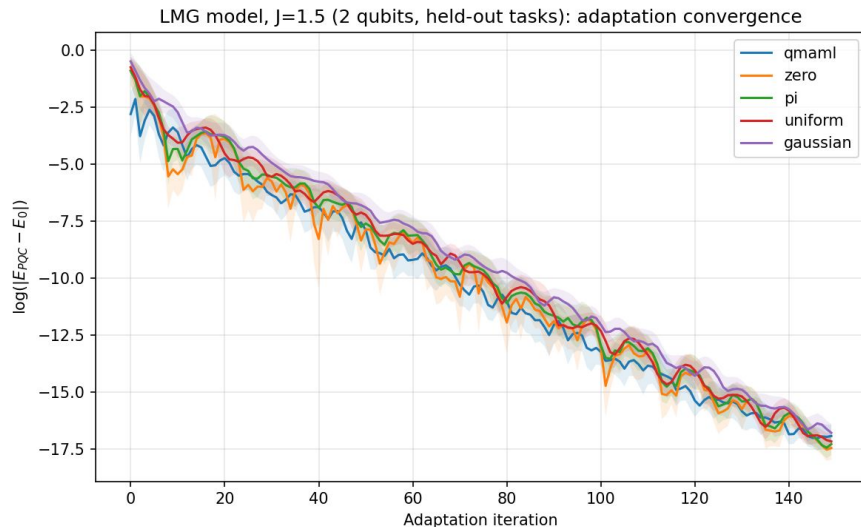
Zero (ties): -18.05

pi (worst): -7.32

Q-MAML ties Zero-init, both at the precision floor. The task range covers near-vacuum ground states, where Zero-init already starts close a real mechanism, not a coincidence.

# Lipkin-Meshkov-Glick Model

2 qubits, nuclear-structure-adjacent, included for completeness



## Key numbers

**All 5 schemes: -16.8 to -17.5 (within noise)**

Q-MAML rank: 4th of 5 - not meaningful

Hilbert space: only 4 states

A genuine null result, kept rather than dropped. No scheme is distinguishable from any other  
the Hilbert space may simply be too small for a bad initialization to get stuck in.

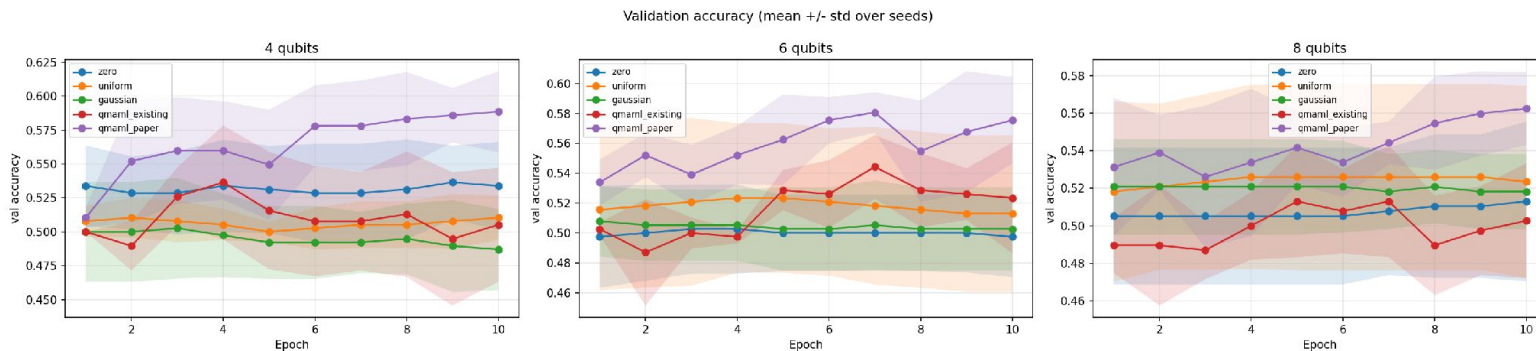
# Ten Tracks, Ten Verdicts

Same data as the individual track slides, side by side.

| Track | Hamiltonian                          | Q | Q-MAML | Best classical    | Verdict |
|-------|--------------------------------------|---|--------|-------------------|---------|
| A     | Schwinger model (L=2)                | 4 | ~0%    | 0.06% (pi)        | Win     |
| B     | Z2 lattice gauge theory              | 8 | 59.16% | 23.07% (uniform)  | Loss    |
| C     | $\phi^4$ scalar field                | 6 | 5.86%  | 5.77% (zero)      | Win     |
| D     | SU(2) lattice gauge theory           | 4 | ~0.00% | 0.02% (pi/gauss.) | Win     |
| E     | Neutrino oscillations (log-gap)      | 4 | -14.35 | -11.39 (pi)       | Win     |
| F     | SUSY quantum mechanics (log-gap)     | 4 | -4.87  | -4.75 (uniform)   | Win     |
| G     | Non-Abelian D8 gauge theory          | 7 | 10.49% | 10.97% (gauss.)   | Win     |
| H     | Nuclear EFT (deuteron/triton/He-3)   | 8 | 0.89%  | 0.89% (all tie)   | Tie     |
| I     | Scalar Yukawa coupling (log-gap)     | 4 | -18.15 | -18.05 (zero)     | Tie     |
| J     | Lipkin-Meshkov-Glick model (log-gap) | 2 | -16.94 | -17.46 (zero)     | Null    |

# Classification, Side by Side

| Qubits | Q-MAML                 | Zero            | Uniform         | Gaussian        | GSoC baseline   |
|--------|------------------------|-----------------|-----------------|-----------------|-----------------|
| 4      | <b>0.589 +/- 0.035</b> | 0.534 +/- 0.037 | 0.510 +/- 0.019 | 0.487 +/- 0.034 | 0.505 +/- 0.049 |
| 6      | <b>0.576 +/- 0.033</b> | 0.497 +/- 0.031 | 0.513 +/- 0.060 | 0.503 +/- 0.032 | 0.523 +/- 0.043 |
| 8      | <b>0.563 +/- 0.023</b> | 0.513 +/- 0.049 | 0.523 +/- 0.059 | 0.518 +/- 0.023 | 0.503 +/- 0.036 |



Significant vs. Gaussian and GSoC baseline at all three qubit counts ( $p < 0.05$ , Welch's t-test,  $n = 4$  seeds). Second dataset (quark-gluon, 2 seeds) replicates the win directionally.

# Key Findings

1

Gauge structure eliminated analytically (A, D, G) consistently favors Q-MAML; penalty-enforced gauge invariance (B) is the one clear loss.

2

Two "ties" (H, I) are real wins a scoreboard would miss -- same final answer, but Q-MAML avoids a shared oscillatory transient or reaches it via a real mechanism.

3

The zero-parameter fixed point is not universal, and not simply about Pauli-sum vs. Hermitian-matrix Hamiltonians (E breaks that link outright).

4

One genuine null result (J) is reported as such -- the benchmark may be too small to be informative, not "Q-MAML fails here."

5

Physics-derived point-estimate initializations are not automatically better than generic random ones (F's basis\_paper, worst of six).

6

Verification catches real bugs before they become wrong results (G's 64-pair representation check).

7

Classification: significant wins on two datasets; the benefit is landscape placement, not barren-plateau resistance.

8

No single classical scheme is a safe default across Hamiltonian families -- Q-MAML nearly is, under interpolation.

# Discussion: Where Next?

Open question for the group:

Shape this toward a real submission (which venue, what timeline) or treat it primarily as GSoC proposal evidence for now?

Concrete next steps, either way:

- Multi-seed replication on the single-seed tracks
- Scale select tracks to match source papers' own sizes (D8 and SU(2) LGT at baryon scale)
- Complete the quark-gluon classification grid to match HIGGS's (seeds x qubit counts)

*Full writeup: 27-page draft + repo with all 18 notebooks, all 10 tracks verified*